

# Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

Wen, Zehai

2026-07-29 22:35:42-04:00

# Chapter 1

## Basic Setup

Given a "high dimensional" data set  $\{\mathbf{x}_i\}_{i=1}^n$  where  $\mathbf{x}_i \in \mathbb{R}^p$  for all  $i = 1, \dots, n$ , the statistician fixes a positive integer  $q < p$  and wishes to find the  $q$  principal components. Assuming that the data is centred (that is, with mean  $\mathbf{0}$ ), the probabilistic principal component analysis model is a statistical model that treats  $\mathbf{x}_i$ 's as realizations of independent random variables  $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$ , where  $\mathbf{W}$  is an unknown  $p \times q$  loading matrix,  $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$  is the latent normal variable, and  $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$  is the Gaussian noise with unknown standard deviation  $\sigma > 0$ , and  $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$  are all independent (for now). Put  $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$  for the unknown parameter to estimate and  $p(\mathbf{x}; \theta)$  for the density of  $\mathbf{X}_1$ , namely the density of  $N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)$ .

We now setup the quotient space structure. Write  $\Theta$  for the collection of all tuples  $(\mathbf{W}, \sigma^2)$  such that  $\mathbf{W}$  is a real  $p \times q$  matrix with  $\text{rank}(\mathbf{W}) = q$  and  $\sigma^2 \in (0, \infty)$ . Given that the collection of all real  $p \times q$  matrix with rank  $q$  is an open subset of the Euclidean space  $\mathbb{R}^{pq}$ , we consider  $\Theta$  as an open subset of the Euclidean space  $\mathbb{R}^{pq+1}$ , hence a Riemannian manifold with the standard Euclidean metric. Next, we define  $(\mathbf{A}, s)$  and  $(\mathbf{B}, t) \in \Theta$  to be equivalent when they give the same density, namely when  $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$ , and write  $(\mathbf{A}, s) \sim (\mathbf{B}, t)$ . The equivalence classes are denoted by  $[(\mathbf{A}, s)]$  or more briefly  $[\mathbf{A}, s]$ . Let us denote by  $O(q)$  the collection of all  $q \times q$  real orthogonal matrices. It is shown in Datta's thesis that  $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$ . Therefore,  $M := \Theta / \sim$  can be identified as the orbits of the right action of  $O(q)$  on  $\Theta$ . Being a smooth, proper, and free action of Lie group on a Riemannian manifold,  $M$  is naturally a Riemannian manifold

### 1.1 Exponential Map and Logarithmic Map Computation

xxx [Nelder and Wedderburn(1972)]

# References

[Nelder and Wedderburn(1972)] J. A. Nelder and R. W. M. Wedderburn. Generalized linear models. *Journal of the Royal Statistical Society. Series A (General)*, 135(3):370–384, 1972. ISSN 00359238, 23972327. URL <http://www.jstor.org/stable/2344614>.